data manipulation 1.py - C:/Users/irsha/AppData/Local/Programs/Python/Python

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```python
import pandas as pd
import numpy as np

# Sample data with missing values
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [25, np.nan, 30, 22],
    'Salary': [50000, 60000, np.nan, 45000]
}

# Create DataFrame
df = pd.DataFrame(data)

# Check missing values
print("Missing Values:")
print(df.isnull().sum())

# Fill missing Age with median
df['Age'] = df['Age'].fillna(df['Age'].median())

# Remove rows where Salary is missing
df = df.dropna(subset=['Salary'])

print("\nCleaned DataFrame:")
print(df)
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/AppData/Local/Programs/Python/Python312/data manipulat
ion 1.py
Missing Values:
Name      0
Age       1
Salary    1
dtype: int64

Cleaned DataFrame:
    Name   Age   Salary
0  Alice  25.0  50000.0
1    Bob  25.0  60000.0
3  David  22.0  45000.0
>>>
```

Ln: 16 Col: 0

data manipulation 4.py - C:/Users/irsha/AppData/Local/Programs/Python/Python

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```python
import pandas as pd

# Sample data
data = {
    'Age': [25, 30, 35, 40, 45],
    'Salary': [50000, 60000, 75000, 90000, 110000],
    'Department': ['HR', 'IT', 'IT', 'Marketing', 'HR']
}

df = pd.DataFrame(data)

# Numeric summary
print("----- Numeric Summary -----")
print(df.describe())

# Categorical summary
print("\n----- Categorical Summary -----")
print(df.describe(include='object'))
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/AppData/Local/Programs/Python/Python312/data manipulat
ion 4.py
----- Numeric Summary -----
             Age         Salary
count   5.000000       5.000000
mean   35.000000   77000.000000
std     7.905694   23874.672773
min    25.000000   50000.000000
25%    30.000000   60000.000000
50%    35.000000   75000.000000
75%    40.000000   90000.000000
max    45.000000  110000.000000

----- Categorical Summary -----

Warning (from warnings module):
  File "C:/Users/irsha/AppData/Local/Programs/Python/Python312/data manipulation
 4.py", line 18
    print(df.describe(include='object'))
Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_
dtypes when 'object' dtype is specified. This behavior is deprecated and will be
 removed in a future version. Explicitly pass 'str' to `include` to select them,
 or to `exclude` to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-mi
gration-select-dtypes for details on how to write code that works with pandas 2
and 3.
       Department
count           5
unique          3
top            HR
freq            2
>>>
```

Ln: 28 Col: 0

data manipulation 5.py - C:/Users/irsha/AppData/Local/Programs/Python/Python312/data...

File Edit Format Run Options Window Help

```python
import pandas as pd

# Sample data
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Eva', 'Frank', 'Grace'],
    'Score': [85, 92, 78, 95, 88, 72, 91]
}

df = pd.DataFrame(data)

# Display first 5 rows
print("Head (Default):")
print(df.head())

# Display first 2 rows
print("\nHead (2 Rows):")
print(df.head(2))

# Display last 3 rows
print("\nTail (3 Rows):")
print(df.tail(3))
```

IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bi
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/AppData/Local/Programs/Python/Python312/data manipula
ion 5.py
Head (Default):
      Name  Score
0    Alice     85
1      Bob     92
2  Charlie     78
3    David     95
4      Eva     88

Head (2 Rows):
    Name  Score
0  Alice     85
1    Bob     92

Tail (3 Rows):
    Name  Score
4    Eva     88
5  Frank     72
6  Grace     91
>>>
```

Ln: 23

data manipulation 2.py - C:/Users/irsha/AppData/Local/Programs/Python/Python312/data...

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```python
import pandas as pd

# Sample data
data = {
    'Employee': ['Amy', 'John', 'Raj', 'Kevin', 'Sora'],
    'Department': ['HR', 'IT', 'IT', 'Sales', 'IT'],
    'Experience': [3, 7, 1, 5, 6]
}

df = pd.DataFrame(data)

# Filter IT employees with experience greater than 5 years
filtered_df = df[(df['Department'] == 'IT') & (df['Experience'] > 5)]

print(filtered_df)
```

IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/AppData/Local/Programs/Python/Python312/data manipulat
ion 2.py
  Employee Department  Experience
1     John         IT           7
4     Sora         IT           6
>>>
```

data manipulation 3.py - C:/Users/irsha/AppData/Local/Programs/Python/Python312/data...

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```python
import pandas as pd

# Sample data
data = {
    'Store': ['North', 'South', 'North', 'West', 'South', 'West'],
    'Sales': [1200, 1500, 900, 1100, 1700, 1300],
    'Employees': [5, 7, 5, 4, 8, 4]
}

df = pd.DataFrame(data)

# Group by Store
summary = df.groupby('Store').agg(
    Total_Sales=('Sales', 'sum'),
    Average_Employees=('Employees', 'mean')
).reset_index()

print(summary)
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.194
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/AppData/Local/Programs/Python/Python312/data m
ion 3.py
   Store  Total_Sales  Average_Employees
0  North         2100                5.0
1  South         3200                7.5
2   West         2400                4.0
>>>
```